

Modular Operator Discovery TSP Report

Overview

- Condition count: 7.
- Best final transfer gap: phase7_full_solver_evolution.
- Best held-out TSPLIB gap: phase7_full_solver_evolution.
- Surviving modular candidates: none.

Run Metadata

- run_name: run_20260519_222129_a
- started_at_local: 2026-05-19 22:21:29
- finished_at_local: 2026-05-19 22:52:43
- duration_hhmm: 00:31
- duration_seconds: 1873.514
- seed_offset: 0
- replicate_label: a
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_baseline_heuristic_only	baseline_only	none	score_only	0.235058	0.054968	0.121317	0.0	0.0	False	0.0	0.0
phase7_full_solver_evolution	full_solver	none	score_only	0.130772	0.003941	0.084652	0.862117	0.76	False	0.0	0.0
phase7_modular_operator_evolution	modular_operator	none	score_only	0.235058	0.046321	0.115856	0.804218	0.445	False	0.008417	0.849429
phase7_modular_operator_random_replay	modular_operator	random	score_only	0.235058	0.054968	0.121317	0.881696	0.48	False	0.003487	0.010929
phase7_modular_operator_diversity_residual_replay	modular_operator	diversity_residual	score_only	0.176662	0.056269	0.100624	0.0	0.4425	False	-0.037302	1.442123

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_modular_operator_compression_pressure	modular_operator	none	novelty_gate	0.235058	0.050827	0.118702	0.691789	0.44	False	0.007173	0.702383
phase7_modular_operator_pareto_selection	modular_operator	none	pareto	0.235871	0.069893	0.131043	0.811695	0.435	False	0.005155	-0.126442

Condition Notes

phase7_baseline_heuristic_only

- Execution mode baseline_only on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.0, complexity 0.0, and adaptation efficiency 0.0.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_full_solver_evolution

- Execution mode full_solver on host scaffold whole_solver.
- Final gaps: TSPLIB 0.130772, family holdout 0.003941, combined 0.084652.
- Accepted novelty 0.862117, complexity 0.76, and adaptation efficiency 0.009542.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_modular_operator_evolution

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.046321, combined 0.115856.
- Accepted novelty 0.804218, complexity 0.445, and adaptation efficiency 0.009096.
- Last-epoch runtime 50.5346 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.008417, positive scaffolds 1, Pareto runtime inflation 0.849429.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.766132, "gap_delta": -0.002615, "runtime_inflation": 0.849429, "same_gap_faster": false}

phase7_modular_operator_random_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.881696, complexity 0.48, and adaptation efficiency 0.006232.
- Last-epoch runtime 56.145825 ms and distance evaluations 3810.75.

- Validation: surviving False, transplant delta 0.003487, positive scaffolds 0, Pareto runtime inflation 0.010929.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.0, "gap_delta": 0.0, "runtime_inflation": 0.010929, "same_gap_faster": false}

phase7_modular_operator_diversity_residual_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.176662, family holdout 0.056269, combined 0.100624.
- Accepted novelty 0.0, complexity 0.4425, and adaptation efficiency 0.0.
- Last-epoch runtime 55.66275 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta -0.037302, positive scaffolds 4, Pareto runtime inflation 1.442123.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 13, "complexity_score": 0.46, "distance_eval_inflation": 11.648842, "gap_delta": -0.019489, "runtime_inflation": 1.442123, "same_gap_faster": false}

phase7_modular_operator_compression_pressure

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.050827, combined 0.118702.
- Accepted novelty 0.691789, complexity 0.44, and adaptation efficiency 0.020983.
- Last-epoch runtime 117.395575 ms and distance evaluations 7522.75.
- Validation: surviving False, transplant delta 0.007173, positive scaffolds 1, Pareto runtime inflation 0.702383.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.788207, "gap_delta": -0.003304, "runtime_inflation": 0.702383, "same_gap_faster": false}

phase7_modular_operator_pareto_selection

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235871, family holdout 0.069893, combined 0.131043.
- Accepted novelty 0.811695, complexity 0.435, and adaptation efficiency 0.00268.
- Last-epoch runtime 198.651025 ms and distance evaluations 7714.75.
- Validation: surviving False, transplant delta 0.005155, positive scaffolds 1, Pareto runtime inflation -0.126442.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": -0.01997, "gap_delta": 0.009726, "runtime_inflation": -0.126442, "same_gap_faster": false}

Judge Appendix

TSP Modular-Operator Discovery Suite Review

Summary of Evaluation Criteria

- **Primary transfer evidence:** held-out TSPLIB gap and family holdout gap
- **Modular operator interest requires:** validation surviving transplant and ablation tests
- **Claims limited to:** reusable operator structure if modular conditions are met

- ****Algorithm discovery claims require:**** operator survival on scaffold and family holdouts

Key Findings

1. Baseline Heuristic Only (phase7_baseline_heuristic_only)

- Heldout TSPLIB gap: 23.51%
- Family gap: 5.5%
- No modular operator, no novelty or survival expected.
- Not a candidate for modular operator claims.

2. Full Solver Evolution (phase7_full_solver_evolution)

- Final TSPLIB gap: 13.07%
- Family gap: 0.39%
- No modular operators extracted or validated.
- Does not provide reusable modular operator structure.

3. Modular Operator: restart_escape_control_2opt_transfer

- Operator type: **restart_controller**
- Validation:
- Ablation shows mild gap differences (~ 0.0026), but disabling operator slightly increases gap $\rightarrow$ marginal ablation support.
- Transplant yields ****mean gap delta $+0.0084$ ****, indicating slight degradation on different scaffolds.
- Only 1 scaffold shows positive transplant effect.
- Transfer and family gaps:
- Family gap reduced by 2.7% on helpful families (clustered_tsp, two_cluster_bottleneck_tsp).
- TSPLIB gap unchanged at 23.5%, no improvement on heldout TSPLIB family.
- Runtime inflation high ($\sim 0.85x$)
- Complexity low (12 lines), compact restart controller mechanism.
- Failure to transplant cleanly to some scaffolds.
- Outcome: ****Does not survive transplant and family holdouts sufficiently****, no surviving candidate.

4. Modular Operator: det_escape_schedule_double_bridge_then_segment_reversal

- Operator type: **perturbation**
- Validation:
- Ablation shows no difference; disabling operator does not hurt performance.
- No positive family gain; no improvement over baseline.

- Transplant mean gap delta +0.0035 (slight degradation).
- No scaffolds with positive transplant gain.
- TSPLIB and family gaps not improved.
- Runtime inflation negligible.
- Outcome: ****No ablation or transplant validation****, not an interesting reusable operator.

5. Modular Operator: **scaffold_selector_structure_aware**

- Operator type: **scaffold_selector**
- Validation:
 - Ablation indicates disabling operator worsens gap by ~ 0.02 , but best variant also shows regression.
 - Some positive family gains (heldout_tsplib, clustered_tsp, two_cluster_bottleneck_tsp).
 - Severe regressions in grid_like_tsp and uniform_euclidean families.
- Transplant mean gap delta is negative (-0.037) indicating some positive transfer but with high runtime inflation ($\sim 1.44x$).
- Positive transplant gains in 4 scaffolds, but large runtime costs and inconsistent performance.
- Outcome: Although partially reusable, inconsistent family and transplant results mean ****does not survive conservative criteria****.

6. Modular Operator: **transfer_restart_controller_bottleneck_escape_dense**

- Operator type: **restart_controller**
- Validation:
 - Ablation shows small gap increase (~ 0.0033) when disabled, indicating a mild effect.
 - Family gains limited to one family (two_cluster_bottleneck_tsp).
- Transplant mean gap delta +0.0072 (degradation), only 1 positive scaffold.
- Runtime inflation moderate ($0.7x$).
- Does not transplant well to some scaffolds.
- Outcome: Moderate effect but ****fails robust transfer and transplant survival****.

7. Modular Operator: **candidate_pruner_scale_by_structure_escape_bias**

- Operator type: **candidate_pruner**
- Validation:
 - Ablation results show disabling reduces gap (~ 0.01), indicating operator may hurt rather than help.
 - No family shows improvement; one family shows severe regression.
- Transplant mean gap delta +0.005 (degradation), only one scaffold positive.
- Runtime slightly reduced but performance worsens overall.
- Outcome: No ablation or transplant validation, ****not reusable or beneficial****.

Overall Conclusion

- ****No modular operator survives transplant and ablation validation robustly enough to be considered reusable.****
- All modular operators fail to reduce held-out TSPLIB gap or family holdout gap in a consistent manner.
- Runtime overhead is significant for some operators with no gap improvements.
- Ablations frequently show operator disabling does not hurt or even improves results.
- Transplant results show mostly neutral or negative gap delta; only isolated positive scaffolds but with tradeoffs.
- Therefore, ****no modular operator supports claims of reusable operator structure or algorithm discovery.****

Recommendations

- Future work should focus on operators that:
- Show significant improvement on heldout TSPLIB and family gaps.
- Pass ablation checks with disabled variants increasing gap substantially.
- Survive transplant tests on multiple scaffolds with neutral or positive gap deltas.
- Maintain manageable runtime overhead and complexity.

Summary Table for Quick Reference

Operator	Operator Type	Heldout Family Gap Δ	Heldout TSPLIB Gap Δ	Ablation Support	Transplant Mean Gap Δ	Runtime Inflation	Survives Modular Validation?
restart_escape_control_2opt_transfer	restart_controller	-0.0275	0.0	Marginal	+0.0084	0.85x	No
det_escape_schedule_double_bridge_then_segment_reversal	perturbation	0.0	0.0	None	+0.0035	0.01x	No
scaffold_selector_structure_aware	scaffold_selector	-0.0072	-0.0583	Weak	-0.0373	1.44x	No
transfer_restart_controller_bottleneck_escape_dense	restart_controller	-0.0035	0.0	Marginal	+0.0072	0.7x	No
candidate_pruner_scale_by_structure_escape_bias	candidate_pruner	+0.0129	+0.0008	None (harmful)	+0.0052	0.87x	No

Final Summary

- ****No evidence of reusable modular operator discovery.****
- ****No modular operator survives scaffold and family transplant tests.****
- ****No algorithm discovery supported.****

- Operators with mild positive family effects do not improve generalization or heldout TSPLIB gap.
- Operators fail ablation or transplant tests critical for claiming reusable modular structure.

Review conducted with conservative interpretation and prioritization of transfer, transplant, ablation, and Pareto tradeoffs per instructions.